

Aflevering 4 - Mikkel Moth Billing

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Afleveringsopgave 4

Vi betragter den harmoniske oscillator i en dimension med notationen som beskrevet i kapitel 2.3. En startbølgefunktion er givet ved $\Psi(x, 0) = \frac{1}{\sqrt{2}}(\psi_n(x) + i\psi_{n+2}(x))$.

- (1) Bestem middelværdien og spredningen af partiklens totale energi.
- (2) Betragt operatoren $\hat{A} = \hat{x}\hat{p} + \hat{p}\hat{x}$. Udtryk denne operator vha. hæve- og sænkeoperatorerne, $\hat{a}_+$ og $\hat{a}_-$.
- (3) Bestem forventningsværdien $\langle \hat{A} \rangle$.

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$$\begin{aligned}\langle \hat{H} \rangle &= \langle E \rangle = \sum |c_n|^2 E_n, \quad E_n = \left(n + \frac{1}{2}\right) \hbar \omega \\ &= \frac{1}{2} \left(n + \frac{1}{2}\right) \hbar \omega + \frac{1}{2} \left(n+2 + \frac{1}{2}\right) \hbar \omega \\ &= \frac{\hbar \omega}{2} (2n+3) \\ &= \hbar \omega \left(n + \frac{3}{2}\right)\end{aligned}$$

$$\sigma = \langle \hat{H}^2 \rangle - \langle \hat{H} \rangle^2 = 0 \quad (2.14)$$

$$\begin{aligned}2 \quad \hat{A} &= \hat{x} \hat{p} + \hat{p} \hat{x} \quad \hat{x} = \sqrt{\frac{\hbar}{2m\omega}} (\hat{a}_+ + \hat{a}_-), \quad \hat{p} = i\sqrt{\frac{\hbar m\omega}{2}} (\hat{a}_+ - \hat{a}_-) \\ &= \sqrt{\frac{\hbar}{2m\omega}} (\hat{a}_+ + \hat{a}_-) i\sqrt{\frac{\hbar m\omega}{2}} (\hat{a}_+ - \hat{a}_-) + i\sqrt{\frac{\hbar m\omega}{2}} (\hat{a}_+ - \hat{a}_-) \sqrt{\frac{\hbar}{2m\omega}} (\hat{a}_+ + \hat{a}_-) \\ &= i\frac{\hbar}{2} \left[(\hat{a}_+ + \hat{a}_-) (\hat{a}_+ - \hat{a}_-) + (\hat{a}_+ - \hat{a}_-) (\hat{a}_+ + \hat{a}_-) \right] \\ &= i\frac{\hbar}{2} \left(\hat{a}_+^2 - \cancel{\hat{a}_+ \hat{a}_-} + \cancel{\hat{a}_- \hat{a}_+} - \hat{a}_-^2 + \hat{a}_+^2 + \cancel{\hat{a}_+ \hat{a}_-} - \cancel{\hat{a}_- \hat{a}_+} + \hat{a}_-^2 \right) \\ &= i\hbar \left(\hat{a}_+^2 - \hat{a}_-^2 \right) ?\end{aligned}$$

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$$3 \quad \langle \hat{A} \rangle = \int_{-\infty}^{\infty} \Psi^* \hat{A} \Psi \, dx$$

$$= i \frac{\hbar}{2} \int_{-\infty}^{\infty} (\psi_n^* - i \psi_{n+2}^*) \hat{A} (\psi_n + i \psi_{n+2}) \, dx$$

Granger ind på alle led i $\hat{A}$ og se om der ikke giver ψ_n eller ψ_{n+2} :

$$= i \frac{\hbar}{2} \int_{-\infty}^{\infty} (\psi_n^* - i \psi_{n+2}^*) (\sqrt{n+1}\sqrt{n+2} \psi_{n+2} + i \sqrt{n+2}\sqrt{n+1} \psi_n) \, dx$$

$$= i \frac{\hbar}{2} (i \sqrt{n+2}\sqrt{n+1} - i \sqrt{n+2}\sqrt{n+1})$$

$$= 0$$

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